

THE COMPLEXITY OF RECOGNIZING MF GROUPS FROM FINITE PRESENTATIONS

SAUERS

A countable group is MF if it admits finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. We show that the set of finite presentations of MF groups is Π_2^0 -complete in the arithmetical hierarchy, and the set of finite presentations of non-MF groups is Σ_2^0 -complete; in particular neither set is recursively enumerable. The upper bound is a description of the MF property by finite certificates. The lower bound encodes the set of programs with infinite domain into finite presentations by an effective Higman embedding and two HNN extensions, using the non-MF group of the companion paper on one branch and, on the other, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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1. INTRODUCTION

Not every countable group is MF: the companion paper *Limits of matrix approximation of groups in operator norm* constructs a finitely generated simple group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ with property (T) such that every homomorphism from H to an MF group is trivial. We determine how hard the MF property is to recognize from a finite presentation, in the Kleene–Mostowski arithmetical hierarchy whose levels Σ_n^0 and Π_n^0 count alternations of quantifiers over a decidable relation.

Theorem A (MF recognition is Π_2^0 -complete). *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is Π_2^0 -complete, and the set of codes of finite presentations of non-MF groups is Σ_2^0 -complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable*

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map $e \mapsto \hat{R}_e$ into finite presentations such that the group presented by $\hat{R}_e$ is MF if and only if the e -th partial computable function has infinite domain.

Proposition 3.3 gives a Π_2^0 description of the MF presentations: a presentation is MF exactly when finite certificates exist at every scale (Section 3). For hardness we reduce the Π_2^0 -complete set of programs with infinite domain to MF recognition: from e we construct a finite presentation $\hat{R}_e$ (Section 5). If the domain is finite, then $H \leq \hat{R}_e$, so $\hat{R}_e$ is not MF. If the domain is infinite, then $\hat{R}_e$ satisfies the hypotheses of Theorem 4.2, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona, and is MF (Sections 4 and 6). The MF property is closed in the space of marked groups (Lemma 3.2). It is also closed under direct limits, by Korchagin [Kor, Proposition 13], so every finitely generated group that is not MF is a quotient of a finitely presented one; Theorem A gives the uniform family and the exact level.

Relation to prior work. Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem A gives the exact level. A classical Π_2^0 -complete recognition problem for finitely presented groups is torsion-freeness, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is Π_2^0 -hard to recognize [BCR]; their construction is the first step of Section 5. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the HNN extensions of Section 5 are chosen so that Theorem 4.2 applies to them.

2. MF GROUPS

A countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch]. Equivalently, G embeds in the unitary group of the

norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\bigoplus_n M_{d_n}(\mathbb{C})$ is the c_0 -direct sum of the sequences with $\|x_n\| \rightarrow 0$. A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, $R = L_{\mathbb{F}_2}(1, 2)$ is the binary Leavitt algebra, the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to $t_i s_j = \delta_{ij}$ and $s_0 t_0 + s_1 t_1 = 1$, and $H = \text{EL}_{12}(R)$ is the subgroup of $\text{GL}_{12}(R)$ generated by the elementary matrices $e_{ij}(a) = 1 + aE_{ij}$, $i \neq j$. By Theorem B of the companion paper, every homomorphism from H to an MF group is trivial; in particular H is not MF.

3. LOCALITY AND FINITE CERTIFICATES

The MF property is defined through sequences of models on all of G , but it is determined by finite data with a fixed separation constant. The reformulation is due to Korchagin [Kor].

The following finite-domain formulation is derived from Korchagin [Kor, Proposition 7].

Lemma 3.1 (locality with fixed separation). *A countable group G is MF if and only if for every finite set $F \subseteq G$ containing 1 and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V: F \rightarrow \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\begin{aligned} \|V(gh) - V(g)V(h)\| &\leq \varepsilon \quad \text{whenever } g, h, gh \in F, \\ \|V(g) - 1\| &\geq 1 \quad \text{for } g \in F \setminus \{1\}. \end{aligned}$$

Proof. If every finite $F \ni 1$ and every $\varepsilon > 0$ admit such a map, choose an exhaustion of G by finite sets F_n closed enough that $g, h, gh \in F_n$ for all large n , take a map for $(F_n, 1/n)$, and extend it by 1 outside F_n ; the resulting sequence is an MF model. Conversely, given MF models V_n , Korchagin [Kor, Proposition 7] amplifies the separation: replacing $V_n(g)$ by the direct sum of its tensor powers $V_n(g)^{\otimes k}$, $k \leq K$, multiplies the defects by at most K and raises $\limsup_n \|V_n(g) - 1\|$ above 1 for each of finitely many g , since an eigenvalue $e^{i\theta}$ of $V_n(g)$ with θ small has a power $e^{ik\theta}$ at angular distance at least $\pi/2$ from 1. A block direct sum of these amplified models over finitely many late indices, one for each $g \in F \setminus \{1\}$, then has defect at most ε on F and separation at least 1 from the identity on $F \setminus \{1\}$. $\square$

Lemma 3.2 (marked closedness). *The class of MF groups is closed in the space of marked groups. In particular, every finitely generated group that is not MF is a quotient of a finitely presented group that is not MF.*

Proof. An (F, ε) -model in the sense of Lemma 3.1 depends only on the multiplication table of F , that is, on which products gh with $g, h \in F$ lie in F and which elements of F are equal. If $G = \langle X \mid N \rangle$ is finitely generated and

not MF, Lemma 3.1 gives a finite $F \ni 1$ and $\varepsilon > 0$ with no (F, ε) -model. Represent the elements of F by words in X ; the finitely many equalities among the words and their pairwise products that hold in G are consequences of finitely many relators $N_0 \subseteq N$, and the inequalities hold in every marked group mapping onto G . Hence every marked group that agrees with G on the finitely many words involved has a multiplication table isomorphic to that of F and is not MF. Thus the MF locus is closed. Applied to $\langle X \mid N_0 \rangle$, which maps onto G and has the same table, it gives the finitely presented quotient. $\square$

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is Σ_1^0 if $A = \{x : \exists y R(x, y)\}$ for a decidable relation R , it is Π_1^0 if its complement is Σ_1^0 , and Σ_2^0 and Π_2^0 are the classes $\{x : \exists y \forall z R(x, y, z)\}$ and $\{x : \forall y \exists z R(x, y, z)\}$ with R decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_P for the group presented by the code P . Let MF_{fp} be the set of codes P such that G_P is MF, and NONMF_{fp} its complement. Korchagin’s Proposition 8 [Kor] is the finite-presentation form of Lemma 3.1; the certificates below add normal-closure witnesses for the trivial words, which is what makes the relation decidable.

Proposition 3.3 (finite certificates). *There is a decidable relation $C(P, n, c)$ on triples of natural numbers such that a code P lies in MF_{fp} if and only if for every n there is c with $C(P, n, c)$. Consequently $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$.*

Proof. Let $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m \rangle$. A certificate at scale n is a triple $c = (d, \ell, \pi)$ consisting of a dimension $d \geq 1$, a labelling ℓ of every reduced word of length at most n in $x_1^{\pm 1}, \dots, x_k^{\pm 1}$ by one of the symbols **T** and **S**, and, for every word w labelled **T**, an expression of w as a product of conjugates of the relators and their inverses that freely reduces to w . Such an expression is checkable, and its existence forces $w = 1$ in G_P . Let $\Phi(P, n, c)$ be the statement that there are $U_1, \dots, U_k \in \mathcal{U}(d)$ with $\|r_i(U) - 1\| \leq 2^{-n}$ for $i \leq m$ and $\|w(U) - 1\| \geq 1/4$ for every word w labelled **S**, where $w(U)$ is the evaluation of w at $U_1, \dots, U_k$ with $x_j^{-1} \mapsto U_j^*$. Writing $U_j = A_j + iB_j$ with real matrices, unitarity is a system of polynomial equations, and a bound $\|X\| \leq t$ or $\|X\| \geq t$ on a matrix whose entries are polynomials in the variables is a semialgebraic condition, since $\|X\| \leq t$ holds precisely when $t^2 - X^*X$ is positive semidefinite. Thus $\Phi(P, n, c)$ is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski’s theorem [Ta]. Let $C(P, n, c)$ hold when c is a well-formed certificate at scale n whose **T**-expressions check and $\Phi(P, n, c)$ holds. Then C is decidable.

Suppose that G_P is MF. Fix n , let $L = \max(n, |r_1|, \dots, |r_m|)$, let F be the set of images in G_P of all words of length at most L , together with

their inverses, and let $\varepsilon = 2^{-n}/(4L)$. Lemma 3.1 gives $V: F \rightarrow \mathcal{U}(d)$ with $V(1) = 1$, defect at most ε on F , and $\|V(g) - 1\| \geq 1$ for $g \in F \setminus \{1\}$. Put $U_j = V(\bar{x}_j)$, where $\bar{w}$ denotes the image of a word w in G_P . For $g \in F$ we have $g^{-1} \in F$ and $\|V(g)V(g^{-1}) - 1\| \leq \varepsilon$, so $\|V(g^{-1}) - V(g)^*\| \leq \varepsilon$. For a word $w = y_1 \cdots y_\ell$ of length $\ell \leq L$, all of whose prefixes have images in F , induction on ℓ gives $\|w(U) - V(\bar{w})\| \leq 2\ell\varepsilon$. In particular $\|r_i(U) - 1\| \leq 2L\varepsilon \leq 2^{-n}$. Label a word w of length at most n by T if $\bar{w} = 1$, with a normal-closure expression as witness, and by S otherwise; for the latter, $\|w(U) - 1\| \geq \|V(\bar{w}) - 1\| - 2n\varepsilon \geq 1 - \frac{1}{2} \geq \frac{1}{4}$. So $C(P, n, c)$ holds for this certificate.

Conversely, suppose that for every n some certificate $c_n = (d_n, \ell_n, \pi_n)$ satisfies $C(P, n, c_n)$, and choose $U^{(n)} \in \mathcal{U}(d_n)^k$ witnessing $\Phi(P, n, c_n)$. For each $g \in G_P$ fix a reduced word w_g representing g , with w_1 the empty word, and put $V_n(g) = w_g(U^{(n)})$. Then $V_n(1) = 1$. For $g, h \in G_P$ the word $t = w_g w_h w_{gh}^{-1}$ is trivial in G_P , so it is freely equal to a product of A conjugates of relators and their inverses, for some A depending on g and h but not on n . Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since $\|xy - 1\| \leq \|x - 1\| + \|y - 1\|$ for unitaries x, y ,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \|t(U^{(n)}) - 1\| \leq A 2^{-n} \rightarrow 0.$$

For $g \neq 1$ the word w_g is nontrivial in G_P , so for $n \geq |w_g|$ the certificate c_n cannot label it T , hence labels it S , and $\|V_n(g) - 1\| \geq 1/4$. Thus $\limsup_n \|V_n(g) - 1\| \geq 1/4 > 0$, and G_P is MF.

The displayed equivalence exhibits MF_{fp} as $\{P : \forall n \exists c C(P, n, c)\}$, a Π_2^0 set, and its complement as a Σ_2^0 set. $\square$

4. HNN EXTENSIONS WITH A CORONA CONJUGATOR

In the companion paper, we prove non-approximability using one-sided conjugation in a norm matrix corona. If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \text{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, while the trace and the left regular representation prove that the group embeds in the resulting algebra.

A countable group G admits a *tracial MF realization* (A, ρ, τ) if A is a separable unital MF C^* -algebra, $\rho: G \rightarrow \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a group embeds in $\mathcal{U}(A)$, so it is MF. Padding blocks with zeros embeds every norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ isometrically into $\prod_n M_n(\mathbb{C})/\bigoplus_n M_n(\mathbb{C})$, so the MF algebras of Section 2 are exactly the separable C^* -algebras that embed in the latter corona, as in [BK, Sh].

Lemma 4.1 (residually finite groups). *Every countable residually finite group admits a tracial MF realization.*

Proof. Let G be countable and residually finite, and choose finite-index normal subgroups $N_1 \geq N_2 \geq \dots$ with $\bigcap_k N_k = \{1\}$; this is possible because G is countable. Let $d_k = [G : N_k]$ and let $\lambda_k: G \rightarrow \mathcal{U}(d_k)$ be the left regular representation of G/N_k . In the corona $\mathcal{Q}_{\mathbf{d}}$ put $\rho(g) = [(\lambda_k(g))_k]$; then ρ is a homomorphism into the unitary group. Fix a free ultrafilter ω on $\mathbb{N}$ and set $\tau([(x_k)_k]) = \lim_{\omega} \text{tr}_{d_k}(x_k)$, where tr_{d_k} is the normalized trace on $M_{d_k}(\mathbb{C})$; this is well defined because $|\text{tr}_{d_k}(x_k)| \leq \|x_k\|$, and it is a tracial state on $\mathcal{Q}_{\mathbf{d}}$. If $g \neq 1$, then $g \notin N_k$ for all large k , so $\text{tr}_{d_k}(\lambda_k(g)) = 0$ for all large k and $\tau(\rho(g)) = 0$. Let A be the C^* -algebra generated by $\rho(G)$ and the unit of $\mathcal{Q}_{\mathbf{d}}$; it is separable, unital, and MF, and $(A, \rho, \tau|_A)$ is a tracial MF realization of G . $\square$

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta: S \rightarrow G$ be an injective homomorphism. The HNN extension

$$(1) \quad R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S) \rangle$$

contains G by Britton's lemma [Bri].

Theorem 4.2 (HNN permanence with a corona conjugator). *In the situation of (1), suppose that G admits a tracial MF realization (A, ρ, τ) , and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$ -homomorphism $\iota: A \rightarrow \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W \iota \rho(s) W^* = \iota \rho(\theta(s)) \quad (s \in S).$$

Then R admits a tracial MF realization. In particular, R is MF.

Proof. Write $D = C^*(\iota \rho(G)) \subseteq \mathcal{Q}$, a separable C^* -algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota \rho(S))$ and $B_1 = C^*(\iota \rho(\theta(S)))$, both containing p . Conjugation by W restricts to a $*$ -isomorphism $\Theta: B_0 \rightarrow B_1$ carrying $\iota \rho(s)$ to $\iota \rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. We write d and u for the images of $d \in D$ and u in U . By construction, U has the following universal property: whenever $\pi: D \rightarrow E$ is a unital $*$ -homomorphism into a unital C^* -algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$ -homomorphism $U \rightarrow E$ extending π and sending u to v . We first embed U in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of R inside U .

Step 1: U embeds in an amalgamated free product. This step follows a corner argument of Ueda [Ue]. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \quad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Write e_{ij} and f_{ij} for the matrix units of A_1 and A_2 . Since $\iota_A(p, 0) = \iota_B(p, 0)$ and $\iota_A(0, p) = \iota_B(0, p)$, we have $e_{11} = f_{11} =: e$ and $e_{22} = f_{22}$ in P . Define $\pi_D: D \rightarrow ePe$ by $\pi_D(d) = \text{diag}(d, 0)$, a unital $*$ -homomorphism into the corner, and put $w = e_{12}f_{21} \in ePe$. Then $ww^* = e_{12}f_{22}e_{21} = e_{12}e_{22}e_{21} = e$ and $w^*w = f_{12}e_{22}f_{21} = f_{12}f_{22}f_{21} = e$, so w is a unitary of ePe . For $b \in B_0$ the element $\text{diag}(b, 0)$ is $\iota_A(b, 0) = \iota_B(b, 0)$, so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}(f_{21}\text{diag}(b, 0)f_{12})e_{21} = e_{12}\iota_B(0, \Theta(b))e_{21} \\ &= e_{12}\iota_A(0, \Theta(b))e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses $\iota_B(0, \Theta(b)) = \text{diag}(0, b)$ in A_2 . The universal property of U gives a $*$ -homomorphism $\Phi: U \rightarrow ePe \subseteq P$ with $\Phi|_D = \pi_D$ and $\Phi(u) = w$. We prove that Φ is injective by constructing $\Psi: P \rightarrow M_2(U)$ with $\Psi\Phi(x) = \text{diag}(x, 0)$. Let $\Psi_1: A_1 \rightarrow M_2(U)$ be the entrywise inclusion $D \subseteq U$, and let $\Psi_2: A_2 \rightarrow M_2(U)$ be the entrywise inclusion $B_0 \subseteq U$ followed by conjugation by $\text{diag}(1, u)$. On C we have $\Psi_1(\iota_A(c_0, c_1)) = \text{diag}(c_0, c_1)$ and $\Psi_2(\iota_B(c_0, c_1)) = \text{diag}(c_0, u\Theta^{-1}(c_1)u^*) = \text{diag}(c_0, c_1)$, so the pair induces a $*$ -homomorphism $\Psi: P \rightarrow M_2(U)$. On generators, $\Psi\Phi(d) = \text{diag}(d, 0)$ and $\Psi\Phi(u) = \Psi_1(e_{12})\Psi_2(f_{21}) = \text{diag}(u, 0)$. Hence $\Psi\Phi$ is the injective $*$ -homomorphism $x \mapsto \text{diag}(x, 0)$ of U into $M_2(U)$, and Φ is injective.

Step 2: P is MF. By Shulman's criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable C^* -algebras is MF as soon as there are injective $*$ -homomorphisms φ_A and φ_B of A_1 and A_2 into one norm matrix corona with $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$. Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A: M_2(D) \rightarrow M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B: M_2(B_0) \rightarrow M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\text{diag}(1, W)$. Both are injective, and on C ,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because Θ is conjugation by W . Hence P is MF, and so are its C^* -subalgebras ePe and $\Phi(U) \cong U$.

Step 3: the tracial state. The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of D for T , the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D , and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G . Thus there is a $*$ -homomorphism $\pi: D \rightarrow C_r^*(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G , so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$ -homomorphism $C_r^*(G) \rightarrow C_r^*(R)$. Composing, we obtain $\sigma_0: D \rightarrow C_r^*(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R , and two $*$ -homomorphisms that

agree on generators agree on B_0 . The universal property of U gives $\sigma: U \rightarrow C_r^*(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j: R \rightarrow \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (1) hold in U , since $u\iota\rho(s)u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^* -algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1 \rangle$ of $C_r^*(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. Hence (A', j, τ') is a tracial MF realization of R . $\square$

Injectivity of j follows from $\sigma \circ j = \lambda_R$, not from the covariant pair $(\iota\rho, W)$, which need not be faithful; in the applications below it is not.

Corollary 4.3 (central HNN extensions). *If G admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t, s] = 1 \ (s \in S) \rangle$ admits a tracial MF realization.*

Proof. Apply Theorem 4.2 with θ the inclusion of S , with ι any injective $*$ -homomorphism of A into a norm matrix corona, and with $W = 1$. $\square$

5. THE COMPLEXITY OF RECOGNIZING MF GROUPS

Write W_e for the domain of the e -th partial computable function and

$$\text{INF} = \{e : W_e \text{ is infinite}\}, \quad \text{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π_2^0 -complete and FIN is Σ_2^0 -complete [So, Chapter IV]. We now combine the non-MF group H from the companion paper with the permanence theorem, Theorem 4.2. We construct a computable map $e \mapsto \hat{R}_e$ into finite presentations such that the presented group contains H when $e \in \text{FIN}$ and satisfies the hypotheses of Theorem 4.2 when $e \in \text{INF}$, so that it is MF if and only if $e \in \text{INF}$. The construction of Bilanovic–Chubb–Roven gives a recursively presented group C_e that is trivial when $e \in \text{INF}$ and contains H when $e \in \text{FIN}$. We embed C_e in a three-generated recursively presented group $Q_e = F/N_e$, use a Mikhailova subgroup to detect N_e inside a finitely presented group, and apply two HNN extensions, in the manner of Higman’s rope trick [Hig], to obtain $\hat{R}_e$. For $e \in \text{FIN}$, the group $\hat{R}_e$ is not MF; for $e \in \text{INF}$, Section 6 proves that it is MF.

A recursive presentation of H .

Lemma 5.1 (a recursive presentation of H). *The group $H = \text{EL}_{12}(R)$ is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

Proof. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k , induction on monomial length shows that $H = \langle Y \rangle$. The monomials in s_0, s_1, t_0, t_1 that contain no subword $t_i s_j$ and no subword $s_1 t_1$ form the standard reduced $\mathbb{F}_2$ -basis of R [Lev]. Indeed, orient the defining relations as

$t_i s_j \rightarrow \delta_{ij}$ and $s_1 t_1 \rightarrow 1 + s_0 t_0$, using a degree-lexicographic order in which $s_1 > s_0$. The critical words are $t_i s_1 t_1$ and $s_1 t_1 s_j$. Their two reductions agree because, for $i, j \in \{0, 1\}$,

$$\delta_{i1} t_1 = t_i + \delta_{i0} t_0, \quad \delta_{1j} s_1 = s_j + \delta_{0j} s_0.$$

The Diamond Lemma [Ber] gives precisely the displayed irreducible monomials as a basis. Products of reduced monomials are computed by the defining reductions, so equality in R is decidable. A word in $Y^{\pm 1}$ is therefore evaluated to an explicit 12×12 matrix over R , and it represents 1 in H exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and $\langle Y \mid \text{that set} \rangle$ is a recursive presentation of H . $\square$

The group C_e .

Lemma 5.2 (the group C_e). *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \text{INF}$ and H embeds in C_e if $e \in \text{FIN}$.*

Proof. Apply the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] to the trivial group and H . Let $\langle Y \mid T \rangle$ be the recursive presentation of H from Lemma 5.1. Take generators $y_{i,\ell}$ for $i \geq 1$ and $\ell \in Y$, one copy Y_i of Y for each i , and as relators all words of T rewritten in each copy Y_i , together with every generator $y_{i,\ell}$ for which $i \leq |W_e|$. The last family is recursively enumerable uniformly in e : enumerate W_e , and whenever its i -th element appears, enumerate all of Y_i . The group presented is the free product of the copies $H_i \cong H$, $i \geq 1$, with the first $|W_e|$ copies killed. If W_e is infinite every copy is killed and C_e is trivial; if W_e is finite, C_e is the free product of the remaining copies and contains H . $\square$

A three-generator embedding. Let $F = F(x, y, t)$ be the free group on x, y, t , and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i \geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$(2) \quad B(C) = \langle C * F(x, y), t \mid t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle.$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$, let $q_+ : F \rightarrow Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

- Lemma 5.3** (the three-generator embedding). (1) $B(C)$ is an HNN extension of $C * F(x, y)$, so C embeds in $B(C)$; it is generated by x, y, t ; and a recursive presentation of $B(C)$ on x, y, t is computable from a recursive presentation of C . We write $B(C) = F/N_C$ and $q_C : F \rightarrow B(C)$ for the quotient map.
- (2) Killing C induces a surjection $B(C) \rightarrow Q_+$; hence $N_C \leq N_+$, with equality when C is trivial.
- (3) The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j : Q_+ \rightarrow P = F(x_1, y) \times F(x_2, t)$. In particular, Q_+ is residually finite.

Proof. (1) The elements y_i , $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x, y)$ [LS, Chapter I]. The retraction $C * F(x, y) \rightarrow F(x, y)$ killing C sends $c_i y_i$ to y_i , so a nontrivial reduced word in the $c_i y_i$ maps to a nontrivial word in the y_i ; hence the $c_i y_i$ freely generate a free subgroup, and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. Thus (2) is an HNN extension, and the base, hence C , embeds by Britton's lemma. The relation $c_i = t y_i t^{-1} y_i^{-1}$ shows that x, y, t generate $B(C)$, and substituting this expression for c_i into the relators of C and into (2) produces a recursive presentation on x, y, t , uniformly in the presentation of C .

(2) The relators of (2) map to the relators $[t, y_i]$ of Q_+ when every c_i is sent to 1, and the map is the identity on x, y, t .

(3) The relators $[t, y_i]$ are sent to $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$, so j is a homomorphism. Let N_0 be the normal closure of $\{y, t\}$ in Q_+ . Then $Q_+/N_0 = \langle x \rangle$ is infinite cyclic and the extension splits, so $Q_+ = N_0 \rtimes \langle x \rangle$. Rewriting the presentation of Q_+ over the transversal $\{x^n\}$ by the Reidemeister-Schreier method [LS, Chapter II] presents N_0 on the generators $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, $n, m \in \mathbb{Z}$, with relators the conjugates $x^m [t, y_n] x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m, y_n]$. Hence $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of y in $F(x_1, y)$, and likewise for t ; so j restricted to N_0 is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1, x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because the exponent sum of x_1 in $j(x^k n)$ is k for $n \in N_0$. Hence j is injective on $N_0 \rtimes \langle x \rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so Q_+ is residually finite. $\square$

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 5.2, and write $q_e: F \rightarrow Q_e$. By Lemmas 5.2 and 5.3, a recursive presentation of Q_e on x, y, t is computable from e ; $N_e \leq N_+$; if $e \in \text{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \text{FIN}$ then $H \leq C_e \leq Q_e$.

Detecting N_e inside a finitely presented group.

Lemma 5.4 (Higman embedding and the Mikhailova subgroup). *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e \rangle$ of a group Λ_e , words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \leq F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \rightarrow \Lambda_e$, and $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \quad i_e(f) = (f, w_f, 1), \quad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e: F \rightarrow K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the

embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of Q_e . Following Mikhailova [Mih], let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z, z) , $z \in X_e$, and $(r, 1)$, $r \in R_e$. If $(u, v) \in M_e$ then $u = v$ in Λ_e , since this holds for the generators. Conversely, if $u = v$ in Λ_e , then uv^{-1} lies in the normal closure of R_e , so it is a product of conjugates $gr^{\pm 1}g^{-1}$ with $r \in R_e$, and $(gr^{\pm 1}g^{-1}, 1) = (g, g)(r^{\pm 1}, 1)(g, g)^{-1} \in M_e$; hence $(u, v) = (uv^{-1}, 1)(v, v) \in M_e$. The first coordinate makes i_e injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in Λ_e , if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. $\square$

The two HNN extensions. Let $P = F(x_1, y) \times F(x_2, t)$ and $j: Q_+ \rightarrow P$ be as in Lemma 5.3, and put

$$K^g = F \times P, \quad L^g = \{(f, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^g, \quad L_e = L_e^0 \times L^g, \quad i: F \rightarrow K_e, \quad i(f) = (i_e(f), (f, 1)).$$

Let

$$(3) \quad \Gamma_e = \langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle, \quad S_e = \langle i(F), v i(F) v^{-1} \rangle \leq \Gamma_e.$$

Lemma 5.5 (the first HNN extension). (1) K_e is a direct product of free groups of finite rank, hence finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.

(2) Γ_e is finitely presented, and a finite presentation is computable from e . The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e , sending the first copy to $i(F)$ and the second to $v i(F) v^{-1}$, is an isomorphism.

(3) There is a homomorphism $\tau_e: S_e \rightarrow Q_e$ with $\tau_e(i(f)) = q_e(f)$ and $\tau_e(v i(f) v^{-1}) = 1$ for $f \in F$.

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 5.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$, so it is finitely generated. Injectivity of i is clear from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f, 1) \in L^g$; the first holds if and only if $f \in N_e$ by Lemma 5.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \leq N_+$, the intersection is $i(N_e)$.

(2) The presentation (3) is finite once $[v, \ell] = 1$ is imposed only for the finitely many generators ℓ of L_e , and it is computable from e because K_e , L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two

copies of F lying outside N_e ; its image is a word $i(f_0) v i(f_1) v^{-1} i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, hence $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define τ_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e , so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with S_e . $\square$

The second HNN extension is

$$(4) \quad R_e = \langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \tau_e(s)) \ (s \in S_e) \rangle,$$

whose edge maps $s \mapsto (s, 1)$ and $s \mapsto (s, \tau_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (4) lists the recursively enumerable relators of Q_e . Its finite replacement is

$$(5) \quad \begin{aligned} \widehat{R}_e = \langle \Gamma_e \times F, u \mid & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(v i(a) v^{-1}, 1)u^{-1} = (v i(a) v^{-1}, 1) \ (a \in \{x, y, t\}) \rangle, \end{aligned}$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 5.5(2), the basis x, y, t of F , and the commutation relators between the two factors.

Lemma 5.6 (the finite presentation). *The groups $\widehat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e . A code of the finite presentation (5) is computable from e .*

Proof. Both maps $f \mapsto u(i(f), 1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \rightarrow \widehat{R}_e$, and they agree on the basis, so $u(i(f), 1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(v i(f) v^{-1}, 1)u^{-1} = (v i(f) v^{-1}, 1)$ for all f . If $n \in N_e$, then $i(n) \in L_e$ by Lemma 5.5(1), so $v i(n) v^{-1} = i(n)$ in Γ_e , and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\widehat{R}_e$. Hence the factor F of $\widehat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (5) become $u(s, 1)u^{-1} = (s, \tau_e(s))$ for s in the generating set $\{i(a), v i(a) v^{-1} : a \in \{x, y, t\}\}$ of S_e ; both sides define homomorphisms on S_e , so the relations hold for all $s \in S_e$. This defines a homomorphism $R_e \rightarrow \widehat{R}_e$; conversely, the relations of (5) hold in R_e after $F \rightarrow Q_e$, because $\tau_e(i(a)) = q_e(a)$ and $\tau_e(v i(a) v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5. $\square$

Lemma 5.7 (the case $e \in \text{FIN}$). *If $e \in \text{FIN}$, then H embeds in $\widehat{R}_e$, and $\widehat{R}_e$ is not MF.*

Proof. For $e \in \text{FIN}$ we have $H \leq C_e \leq Q_e \leq \Gamma_e \times Q_e \leq R_e \cong \widehat{R}_e$ by Lemmas 5.2, 5.3(1), 5.6 and Britton's lemma for (4). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and H is not MF by Theorem B of the companion paper. $\square$

6. THE CASE $e \in \text{INF}$

We now prove that $\widehat{R}_e$ is MF for $e \in \text{INF}$ and deduce Theorem A. Its tracial MF realization lies in a norm reduced product of MF algebras. For C^* -algebras B_n , $n \geq 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \rightarrow 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 6.1 (reduced products of MF algebras). *If every B_n is MF, then every separable C^* -subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

Proof. The second assertion holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k . For the first, let A be a separable C^* -subalgebra, and choose a sequence $(a_i)_{i \geq 1}$ dense in A , with lifts $(a_i^{(n)})_n$. Choose for each n an injective $*$ -homomorphism η_n of B_n into a norm matrix corona, and lifts $(a_i^{(n,k)})_k$ of $\eta_n(a_i^{(n)})$ with $\sup_k \|a_i^{(n,k)}\| \leq \|a_i^{(n)}\| + 1$. Enumerate the noncommutative $*$ -polynomials with coefficients in $\mathbb{Q}(i)$ as $p_1, p_2, \dots$, and let $\mathcal{P}_n$ be the set of those among $p_1, \dots, p_n$ that involve only $a_1, \dots, a_n$, choosing $p_1 = 0$ so that $\mathcal{P}_n$ is nonempty. For $p \in \mathcal{P}_n$ the number $\|p(a^{(n)})\|$ equals $\limsup_k \|p(a^{(n,k)})\|$, since η_n is isometric. Hence there is K_n such that $\|p(a^{(n,k)})\| \leq \|p(a^{(n)})\| + 1/n$ for all $k \geq K_n$ and $p \in \mathcal{P}_n$, and for each $p \in \mathcal{P}_n$ there is $k_p \geq K_n$ with $\|p(a^{(n,k_p)})\| \geq \|p(a^{(n)})\| - 1/n$. Put

$$b_i^{(n)} = \bigoplus_{p \in \mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For $p \in \mathcal{P}_n$ the norm of $p(b^{(n)})$ is the maximum of $\|p(a^{(n,k_{p'})})\|$ over $p' \in \mathcal{P}_n$, which lies between $\|p(a^{(n)})\| - 1/n$ and $\|p(a^{(n)})\| + 1/n$. Every polynomial belongs to $\mathcal{P}_n$ for all large n , so $\limsup_n \|p(b^{(n)})\| = \limsup_n \|p(a^{(n)})\| = \|p(a)\|$ for every p . Therefore $a_i \mapsto [(b_i^{(n)})_n]$ is well defined and isometric on the $*$ -algebra over $\mathbb{Q}(i)$ generated by the a_i , which is dense in A , and it extends to an isometric $*$ -homomorphism of A into a norm matrix corona. $\square$

Lemma 6.2 (tensor synchronization). *Let Γ be a countable group with a tracial MF realization (A_1, ρ_1, τ_1) , let Q be a countable group with homomorphisms $\beta_n: Q \rightarrow B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n , let $S \leq \Gamma$, and let $\tau: S \rightarrow Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n: \Gamma \rightarrow G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^ -algebra A , a homomorphism $V: \Gamma \times Q \rightarrow \mathcal{U}(A)$, a tracial state T on A with $T(V(g, q)) = 0$ for $(g, q) \neq (1, 1)$, and*

a unitary $W \in A$ with $WV(s, 1)W^* = V(s, \tau(s))$ for all $s \in S$. Consequently

$$\langle \Gamma \times Q, u \mid u(s, 1)u^{-1} = (s, \tau(s)) \ (s \in S) \rangle$$

admits a tracial MF realization, and in particular is MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n): \Gamma \times Q \rightarrow G_n \times B_n$, a finite group of order k_n , and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 6.1, let $E = \prod_n B'_n / \bigoplus_n B'_n$, and define

$$V(g, q) = [(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q)))_n], \quad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \text{tr}_{k_n})(x_n),$$

for a free ultrafilter ω . Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \text{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g, q)) = 0$; if $g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n , the regular representation of a finite group has trace zero off the identity, and again $T(V(1, q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n \tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. Their images are therefore isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n \tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n \tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s, 1)W^* = V(s, \tau(s))$ for $s \in S$. Let A be the C^* -algebra generated by $V(\Gamma \times Q)$ and W ; it is separable and unital, and MF by Lemma 6.1. The triple $(A, V, T|_A)$ is a tracial MF realization of $\Gamma \times Q$. For the last assertion, apply Theorem 4.2 to this realization. Let ι be an injective $*$ -homomorphism of A into a norm matrix corona and put $p = \iota(1)$. The element $\tilde{W} = \iota(W) + (1 - p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$, so it is the required conjugator. $\square$

Lemma 6.3 (the case $e \in \text{INF}$). *If $e \in \text{INF}$, then $\hat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

Proof. Let $e \in \text{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\tau_e(i(f)) = q_+(f)$, $\tau_e(v i(f) v^{-1}) = 1$. By Lemma 5.6 it suffices to show that R_e admits a tracial MF realization, and by Lemma 6.2 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \tau_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 5.5(1), hence admits a tracial MF realization by Lemma 4.1, and Γ_e is the central HNN extension of K_e over L_e , hence admits one by Corollary 4.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \dots$ of P with trivial intersection, let $r_n: P \rightarrow C_n = P/P_n$, and put $\beta_n = r_n \circ j: Q_+ \rightarrow C_n$ with j the embedding of Lemma 5.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n .

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f), r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (3): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, j q_+(f))$ of L^g is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ . Hence $\lambda_n: \Gamma_e \rightarrow G_n$ is a homomorphism, and on the generators of S_e ,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \quad \lambda_n(v i(f) v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1: S_e \rightarrow Q_+$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 5.5(3); thus $\pi_0 = \tau_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \tau_e),$$

which is the hypothesis of Lemma 6.2. $\square$

Proof of Theorem A. By Lemma 5.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 6.3 the presented group is MF for $e \in \text{INF}$, and by Lemma 5.7 it is not MF for $e \in \text{FIN}$. Hence INF reduces to MF_{fp} and FIN reduces to NONMF_{fp} under computable many-one reductions. Since INF is Π_2^0 -complete and FIN is Σ_2^0 -complete, and since $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ by Proposition 3.3, the two sets are complete at their levels. A Π_2^0 -complete set is not Σ_2^0 and a Σ_2^0 -complete set is not Π_2^0 , so neither set is recursively enumerable, and in particular neither is decidable. $\square$

The homomorphism V of Lemma 6.2 does not extend faithfully to R_e in the corona: the finite maps λ_n are trivial on the Mikhailova factor, so W commutes with $V(\gamma, 1)$ for every $\gamma \in K_e^0$, although u does not commute with $(\gamma, 1)$ in R_e . Injectivity is obtained in Theorem 4.2 from the universal algebra. We do not know whether the models of $\widehat{R}_e$ can be chosen to converge to $C_r^*(\widehat{R}_e)$ in operator norm, so that the reduced C^* -algebra itself is MF; the argument produces only a tracial state on an abstract MF algebra. We do not know whether $\widehat{R}_e$ is hyperlinear for $e \in \text{INF}$, which would place the recognition problem for hyperlinearity at the same level. Finally, we do not know whether an analogous family of finitely presented groups exists for soficity. Such a construction would require a permutation-model replacement for Lemma 6.2.

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USE OF AI AND FORMAL METHODS

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